

Per-ID Localization FLR: Opposite-Class Target-Decoy Model

Target-decoy on site-determining evidence; per-ID FLR

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1. Goal

Give every $C(c,n) > 1$ peptidoform a **per-ID localization FLR** (a localization q-value: the estimated probability its site assignment is wrong), computed by an LGBM model calibrated against positional- isomer decoys. $C(c,n) \leq 1$ peptidoforms are unambiguous and carry no localization FLR. No single global target FLR is imposed; the per-ID value is reported and users filter (typical phospho operating point ~5-10%).

2. The two scores (recap)

For a (`precursor_idx`, `scan_idx`) instance, using that scan's observed spectrum and the SAME `permute_fragment_mz` the decoy generator uses (full recipe in `site_determining_ion_score.md` and `consistent_target_decoy_scoring.md`):

- Compare the instance's assigned placement to a **reference class** of positional isomers of its sequence (all such **same-composition** library isomers, not just deconvolution candidates and not just single-site moves - no lambda dependency).
- **Distinguishing ions are general (competitors may differ at >1 site)**. A fragment distinguishes isomers A and B iff their **cumulative modification content differs at that cleavage** – $(\# \text{ A-mods at pos } \leq i) \neq (\# \text{ B-mods } \leq i)$. Then A and B predict **different m/z** for it, offset by $(\text{Delta count}) * \text{mod_mass}$, which can be $\pm 1x$, $\pm 2x$, ... the mod mass (where mods pile up before a cleavage). $\text{support}(A)$ = observed intensity at A's own predicted m/z, $\text{support}(B)$ at B's; $s(A, B) = \text{support}(A) / (\text{support}(A) + \text{support}(B))$ on presence/absence weighted by predicted intensity (neutral-loss channels excluded). This replaces the "cleavage between two sites a and a'" special case (which only held for single-site differences).
- The distinguishing-fragment set is computed **per competitor pair** (it can be non-contiguous and multi-magnitude, e.g. {S1,S7} vs {S4,S10} distinguishes on [1,4) and [7,10) separately; {S1,S4} vs {S7,S10} gives a +2x shift on [4,7)). So a distinguishing ion is NOT attributable to one adjacent-site "gap" – match each isomer at its own m/z, don't bin by fixed gaps.
- Aggregate by **MIN over all competitors** (must beat the hardest one; the hardest need not be a single-site move – a 2-site-different isomer whose distinguishing ions are all low-predicted-intensity can be the min, so enumerate every same-composition isomer). Cost is $O(L)$ per pair.

The score (rule: **the decoy must model a *mis-localized* target, which faces its truth** - see 2b(i)):

- **Target S_vs_targets** = min over the **other** target isomers (drops itself), **n-1** competitors. It is scored only against *real* siblings – never against a decoy – so its confidence never rides on where a random shadow landed.
- **Decoy S_vs_targets** = min over **all** target isomers **including its parent**, **n** competitors. The parent is (usually) the present/true isomer, so keeping it makes the decoy get killed by the truth – exactly like a mis-localized target – rather than escape.
- **S_vs_decoys** (optional secondary feature, target instances only) = min over the **decoy** isomers. Not required for the model; **S_vs_targets** is the axis the classifier trains and applies on.

Plus supporting scalars per reference: **n_distinguishing_ions**, **distinguishing_capacity** (total *predicted* distinguishing intensity vs the hardest competitor - the localizability ceiling), **coverage**, **min_pair_capacity** (predicted distinguishing intensity against the single hardest competitor; this is how adjacent/near-tied isomers earn their difficulty rather than being flagged).

2b. Two competitor-set details (worked examples)

(i) Keep the parent for decoys – a decoy must model a *mis-localized* target

The reference for a decoy is not a correctly-localized target but a **mis-localized** one (both are wrong placements), and a mis-localized target **faces its truth**:

- Correct target A (truth A): vs {B, C} – truth is itself, excluded -> wins -> **high**.
- Mis-localized target A (truth B): vs {B, C} – truth **B is in the set** -> killed by B -> **low**.
- Decoy A' (truth = present isomer = its parent A): to match the mis-localized target it must **face its truth** -> **keep the parent**, A' vs {A, B, C} -> killed by A -> **low**.

So the decoy lands on the mis-localized target, not on a correct one. Dropping the parent would instead make A' mirror a *correct* target (it would escape to high) – the wrong reference.

Worked example. Peptide **S A S A S A K**, acceptors **S1, S3, S5** -> targets A=pS1, B=pS3, C=pS5; shadow sites 2/4/6 -> decoys A' (mod->pos2), etc. **The scan contains A (pS1)**, so observed b-ions **b1..b6** are all shifted (+80). A **b_i** is shifted iff mod position <= i:

isomer (mod pos)	b1	b2	b3	b4	b5	b6
A (1)	+	+	+	+	+	+
A' (2)	-	+	+	+	+	+
B (3)	-	-	+	+	+	+
C (5)	-	-	-	-	+	+

- A (target, vs {B,C}): min = **1.0** -> correctly localized vs the other acceptors.
- A' (decoy, vs {A,B,C}, parent kept): **s(A',A)** distinguishes on **b1** (A' unmod at pos 2, A shifted at pos 1; observed **b1** shifted -> supports A) = 0.0. min = **0.0** -> A' rejected, landing on top of a mis-localized target (also 0).

A and A' differ only at the shadow move (site 1 vs 2), so the one ion that rejects A' is **b1**, which lives in the A-vs-A' comparison – keeping the parent is what supplies it. And unlike the target's

competitor set, this does NOT add variance to any *target's* score (a target is only ever scored against real siblings), so a correct target's confidence never depends on where a random shadow fell.

Two residual points, neither fixed by the competitor set: (1) the decoy faces one *extra* non-truth competitor a mis-localized target doesn't, but the **min** is dominated by the truth-kill, so this is a small, at-most-mildly-optimistic bias. (2) The shadow is a **non-acceptor**, so **A'** is killed by an easy, implausible **b1**, whereas a real wrong-*acceptor* error is killed by ions between two real (often farther-apart) sites – so the shadow decoy is *too easy* and can under-estimate the FLR. **The fix is decoy construction (short-distance-biased shadow sites near real acceptors), not C(c,n)-manipulation** (bumping decoy **C** would strand the C=2 stratum with no null). Both are measured on the standards (Section 5).

(ii) Accumulate evidence per site, not just per competitor pair

When competitors differ at more than one site, a single pooled **s(A,B)** can be carried entirely by one site's ions while the other is unconfirmed. Example: peptide **S A A S A A S A A R** (acceptors **S1,S4,S7**), **A={pS1,pS4}** vs **B={pS4,pS7}** (differ at **S1** and **S7**):

- **b-ions** **b1..b6** are +80 heavier for **A** -> they confirm the **S1** end (**A** has **pS1**, **B** does not).
- **y-ions** **y4..** are +80 heavier for **B** -> they confirm the **S7** end.

The two ends are probed by *independent* ion series. If we only observed b-ions, a pooled **s(A,B)** would read high (**S1** confirmed) while **S7** is **entirely unconfirmed** – a false-confident call.

Fix: accumulate **per assigned site s**. For each modified site, gather the ions whose m/z depends on **s's** occupancy (moving the mod off **s** changes them) and compute **s_site(s)** = “mod at **s**” vs “mod moved off **s**”. Peptidoform confidence = **min over its sites** (weakest link); the per-site vector is the informative output (and matches per-site-probability reporting conventions).

- A **multi-site-different competitor contributes to multiple sites** (its b-ions to **S1**, its y-ions to **S7**).
- **Joint-only ions** – whose m/z changes only under a *combined* move (e.g. a +2x fragment where two mods pile up before the cleavage) – confirm the *arrangement* but neither site individually; a site confirmed only by joint ions is flagged **not independently localized**.

New features: **s_site** per site -> **min_site_support** (= the peptidoform localization), **n_sites_independently_confirmed**, **n_sites_joint_only**.

3. The model: target-decoy classifier on **S_vs_targets**

Every instance is scored **S_vs_targets** (2b(i)): a **target** against its **n-1** real siblings, a **decoy** against **all** targets including its parent. The parent (usually the present/true isomer) kills the decoy, so a decoy lands exactly where a **mis-localized target** does – which is precisely what it must model.

- **Positives:** all **target** positional isomers, **S_vs_targets** (vs siblings).
- **Negatives:** all **decoy** positional isomers, **S_vs_targets** (vs all targets, parent kept).
- **Apply:** score each target's **S_vs_targets** and read off the model probability.

A correctly-localized target beats its siblings -> high; a mis-localized target loses to the truly-present sibling -> low, landing in the decoy region; the decoys calibrate that region. The apply

target and the mis-localized target it might be are both scored against real siblings, and the decoy (killed by its truth) sits on the mis-localized target – so the null is the right one. (The decoy’s one extra non-truth competitor is a small, at-most-mildly-optimistic bias; 2b(i).)

Feature blocks. Prior (P: $C(c,n)$, $\log C$, site counts, distances, terminal flags) and competition (W: `iso_group_size_at_scan` = number of competitors in the scan, `weight`, `weight_ratio`, `iso_weight_fraction`, `mbr`) are intrinsic to the (instance, scan). The **S block** is the `S_vs_targets` evidence, now **per-site** (2b(ii)): `min_site_support`, `s_site` distribution, `n_sites_independently_confirmed`, `n_sites_joint_only`, plus `n_distinguishing_ions`, `distinguishing_capacity`, `coverage`, `min_pair_capacity`.

One tradeoff to note (label noise). Because some real (yeast) targets are themselves mis-localized and we don’t know which, a small fraction of the positive labels are “wrong-but-labeled-target.” This is the standard target-decoy situation (a few mislocalized positives; the decoys still define the null) and is tolerable when correct localizations dominate; we monitor it via the standards. (An earlier draft avoided this by scoring targets against decoys at train time – the “opposite-class” variant – but it does not model the mis-localized-target reference cleanly, so we drop it in favor of this target-decoy classifier.)

4. Per-ID localization FLR

The model outputs a score per instance. Calibrate to a per-ID FLR by target-decoy competition on the score:

- Score all instances (targets and decoys alike) on `S_vs_targets`; the decoys are the null.
- Sort by score desc; at score `s`, $FLR(s) = D(>=s) / R(>=s)$ ($C>1$ only).
- Each real ID’s **localization FLR** = **min FLR over cutoffs at or below its score** (monotone q-value).
- **MBR instances are included** on both sides (their decoys are transferred too) -> held to the same standard, no special-casing. Report with-MBR and MBR-excluded columns.
- $C \leq 1$: localization FLR = N/A (unambiguous); still subject to the 1% ID-FDR.

5. Validation on the standards (ground truth, no construction)

The standards are $C>1$ peptidoforms in the library with **known true sites**; they get a reported localization and a per-ID FLR like everything else. So:

1. **Correctness label** per detected standard: is the reported isomer’s site == the true site (`standards_truth`)? This yields the standards’ real **acceptor-site** error rate directly - exactly the error mode we care about.
2. **Calibration curve:** bin detected standards by predicted per-ID FLR; the empirical error rate in each bin should equal the bin’s predicted FLR. Calibrated -> the shadow-decoy-based FLR is accurate on real localization. Predicted < empirical -> decoys are too easy (under-count) -> the short- distance-biased decoy fix (Section 8) is warranted. Predicted > empirical -> conservative (safe), consistent with prior runs (decoy ~9.6% vs empirical ~4%).
3. **Ablation:** correctly-localized standard IDs at matched FLR for prior-only vs +W vs +S, to show the S block earns its cost.

This is the whole test - no deliberately-mislocalized peptides. The standards’ own mis-calls (search

picks the wrong real isomer) are the real wrong-acceptor errors, and the per-ID FLR either predicts their rate or it doesn't.

6. What Phase 1 emits (so we can choose empirically)

The extractor (analysis-only, on the existing combined-search results; reuses `site_determining.jl`) emits, per (`precursor_idx`, `scan_idx`) $C>1$ instance, the per-site **S_vs_targets** block (parent dropped for decoys) plus P and W. It also emits the optional **S_vs_decoys** block cheaply (pairwise **s** is already computed) so we can test whether it adds signal, but the model is trained and applied on **S_vs_targets** alone. Everything is validated by whether the per-ID FLR tracks the standards' empirical error.

7. Phased plan

Phase	Deliverable	Cost	Gate
1	per-site S_vs_targets extractor -> per-instance table (P, W, S; reals+decoys)	analysis-only	do S features separate reals from decoys in high- C /singleton strata?
2	LGBM (target-decoy on S_vs_targets , parent kept for decoys) + per-ID FLR + standards calibration + ablation	analysis-only	does per-ID FLR track standards' empirical error, and beat prior/W-only?
3	Wire per-ID localization FLR into reporting (+ tests)	code + tests	holds within C strata; MBR handled
4 (opt)	Decoy realism (short-distance-biased shadow sites; decoy iRT perturbation) if under-calibrated	generator + rebuild	predicted FLR tracks empirical more tightly

8. Open questions

1. **Multi-mod** (di-phospho, phospho+ox): *subsumed* by the general cumulative-mod-content definition in Section 2 - the competitor set is simply all same-composition isomers, and distinguishing ions fall out of the per-pair content difference (no single-move restriction). Remaining question is only cost control when $C(c,n)$ is large (cap the competitor set by predicted plausibility?).
2. **Separate biological-ambiguity number?** **S_vs_targets** blends “distinguished from real siblings” with the FLR calibration. Do we also surface a standalone “confidently a phosphopeptide but ambiguous between S4/S7” flag (min vs real siblings only)?

3. **Competitor scope:** all library isomers vs RT/weight-plausibility-gated. All-isomers is cleanest (no lambda dependency); gating trades that for less noise from isomers that could never be present.
4. **q-value vs PEP** for the per-ID value (or both).

*Companion: `consistent_target_decoy_scoring.{md,pdf}` (scores + worked example),
 `site_determining_ion_score.{md,pdf}` (S recipe), `localization_scoring_plan.{md,pdf}`
(broader plan), `FINDINGS.md` (empirical basis).*